Multitime memory beyond the quantum regression theorem in sequential measurement statistics

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We investigate the presence of memory in the sequential measurement statistics of an open quantum system, as witnessed by the departure from the quantum regression theorem (QRT), that is, the possibility to predict multitime probabilities from the one-time reduced dynamical map. For factorized initial states, we identify an exact decomposition of the two-time propagator into a QRT-like contribution, fully determined by the reduced dynamical map, and a memory term encoding system–environment correlations across the intervention; in the weak-coupling regime, the memory term yields an explicit second-order correction expressed in terms of the reduced map and bath correlation functions. Furthermore, we introduce an operational quantifier of QRT violations based on the distance between exact and QRT-predicted joint probabilities. Benchmarking the framework on a spin–boson model and using a pseudomode embedding as nonperturbative reference, we comprehensively analyze the impact of spectral-density parameters, environmental temperature, and measurement protocols on the non-Markovianity of the multitime statistics. Comparison with a one-time quantifier shows that reduced-state non-Markovianity and multitime memory are related but inequivalent: the latter, as probed through sequential statistics, is intrinsically protocol dependent and can become visible at higher temporal order even when two-time statistics remain compatible with QRT predictions.

I. INTRODUCTION

In many experimental settings involving open quantum systems, the quantities of interest are inherently multitime: they arise, for example, from control protocols, response and fluctuation functions, photon-counting statistics, and time-resolved spectroscopy [1–6]. In these cases, the relevant information is contained in joint probability distributions or correlators associated with interventions performed at different times, and the reduced dynamical map, which fully characterizes the evolution of the system state at a single time, is in general not enough to determine such multitime quantities.

A standard construction to deal with multitime quantities is provided by the quantum regression theorem (QRT) [7, 8], which expresses them by combining the same reduced dynamical map that describes one-time expectation values. This procedure is known to be valid under the weak-coupling and singular-coupling regime [9] and, more generally, when system–environment correlations generated during the evolution do not affect the subsequent dynamics relevant for the considered quantities [10]. However, finite bath correlation times, structured environments, strong coupling, and correlations built up prior to intermediate interventions can all lead to deviations from the QRT predictions, which calls for dedicated strategies to describe multitime objects in generic regimes [11–17].

Violations of the QRT can be associated to a notion of multitime non-Markovianity, in a sense close to the classical definition of non-Markovian stochastic processes, as the whole hierarchy of joint probability distributions cannot be reconstructed from one-time probabilities and propagators alone. This idea can be formalized by means of quantum stochastic processes [18, 19] and process tensors [20–24], where multitime statistics associated with sequences of interventions are treated as fundamental objects and Markovianity corresponds to a suitable factorization of the multitime correlations. These multitime notions of memory are related to, but distinct [25–28] from, definitions of non-Markovianity based on the reduced dynamics. The latter are typically formulated in terms of properties of the dynamical map, such as P- or CP-divisibility [29–33], and characterize memory effects at the level of the system states. By contrast, QRT violations and, more generally, quantum-stochastic-processes and process-tensor approaches are defined in terms of sequential measurements and joint outcome distributions, so that they provide an operational notion of memory that is not fully determined by one-time properties of the reduced evolution. Recent studies of spin-boson dynamics have shown that single-intervention process tensors can diagnose quantum memory more informatively than reduced-map criteria alone [34], while a complementary line of work relates violations of QRT in two-point correlations to quantum features in the non-Markovian evolution, including system-environment entanglement [35].

In this work, we investigate multitime memory, as witnessed by the breakdown of QRT in sequential measurement statistics, for fixed projective measurements. First, using projection-operator techniques adapted to dynamics with intermediate interventions [12], we derive an exact decomposition of the two-time propagator into a QRT-like term, fully determined by the reduced dynamical map, and a complementary contribution that fully encodes deviations from the QRT into system–environment correlations across the intervention. In the weak-coupling regime, this memory contribution can be expanded perturbatively, leading to an explicit second-order correction expressed in terms of the reduced map and bath correlation functions. On the operational side, we formulate the problem in terms of quantum instruments and introduce a quantifier of QRT breakdown based on the Kolmogorov distance between exact and QRT joint probabilities.

As a benchmark, we apply this framework to the spin–boson model with a Lorentzian spectral density, using

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the pseudomode embedding as a nonperturbative reference [36, 37]. This allows us to assess the accuracy of the perturbative correction and to analyze the dependence of QRT deviations on time, model parameters, temperature, and measurement protocols. We further compare the resulting multitime characterization with a one-time witness of P-divisibility reconstructed from the same reduced dynamics. The comparison shows that one-time and multitime notions capture different aspects of memory in the evolution: while they are both enhanced by a stronger coupling leading to large system-environment correlations, multitime memory depends on the measurement protocol and may become visible at higher temporal order even when lower-order quantities remain close to the corresponding Markovian predictions.

The remainder of the paper is organized as follows. In Sec. II we introduce the open quantum system framework and the description of multitime objects therein, corresponding to correlators and sequential-measurement joint probability distributions. In Sec. III we present an exact decomposition of the two-time propagator into a QRT contribution and a correction, which we explicitly work out at leading order in the coupling strength. In Sec. IV we apply our results to the paradigmatic spin–boson model, quantifying the breakdown of the QRT description, across different dynamical regimes. In particular, we compare QRT violations with a one-time divisibility-based witness of non-Markovianity, and analyze protocol dependence, temperature effects, and higher-order temporal statistics. Section V contains the conclusions and future perspectives of our work.

II. OPEN QUANTUM DYNAMICS

A. System–bath dynamics

We consider a system S with Hilbert space $\mathcal{H}_S$ coupled to an environment (bath) B with Hilbert space $\mathcal{H}_B$, such that the total space is $\mathcal{H} = \mathcal{H}_S \otimes \mathcal{H}_B$. The joint system-environment state $\rho(t)$ evolves according to the Liouville–von Neumann equation [3]

$$\frac{d}{dt}\rho(t) = \mathcal{L}[\rho(t)], \qquad \mathcal{L} = \mathcal{L}_S + \mathcal{L}_B + \mathcal{L}_{SB}, \tag{1}$$

where $\mathcal{L}_X[\rho] = -i[H_X, \rho]$ for $X \in \{S, B, SB\}$ and H_S (H_B) describes the free system (bath) Hamiltonian, while H_{SB} describes the interaction term (units $\hbar = 1$); indeed, $\mathcal{L}$ denotes the Liouvillian on the global $S-B$ system. We also use the free Liouvillian $\mathcal{L}_0 := \mathcal{L}_S + \mathcal{L}_B$ and the interaction-picture coupling superoperator

$$\mathcal{L}_I(t) := e^{-t\mathcal{L}_0}\,\mathcal{L}_{SB}\,e^{t\mathcal{L}_0}. \tag{2}$$

Throughout we assume a coupling that can be written in the form

$$H_{SB} = \lambda \sum_a S_a \otimes B_a, \tag{3}$$

where λ is the coupling-strength parameter. We fix a bath reference state R satisfying

$$\mathcal{L}_B R = 0, \tag{4}$$

i.e., R is stationary under the free bath dynamics. Without loss of generality, we take the bath interaction operators to have vanishing mean in R,

$$\mathrm{Tr}_B[B_a R] = 0 \qquad \forall a, \tag{5}$$

since any nonzero mean can be absorbed into a renormalization of H_S [5]. Further assuming that the bath is Gaussian, its influence on the open-system dynamics is fully enclosed in the correlation functions

$$C_{ab}(t) := \mathrm{Tr}_B[B_a(t)\,B_b\,R]\,, \quad B_a(t) := e^{-t\mathcal{L}_B}[B_a], \tag{6}$$

which are referred to the environmental interaction operators evolving under the free bath dynamics.

The reduced state associated with the open system is obtained via the partial trace, $\rho_S(t) = \mathrm{Tr}_B[\rho(t)]$. In the following we will consider factorized initial states of the form

$$\rho(0) = \rho_S(0) \otimes R, \tag{7}$$

so that we can define the (one-time) reduced dynamical map $\Lambda_S(t)$, $t \geq 0$, by

$$\rho_S(t) = \Lambda_S(t)[\rho_S(0)] := \mathrm{Tr}_B\Big[e^{t\mathcal{L}}[\rho_S(0) \otimes R]\Big], \tag{8}$$

which is a completely positive and trace-preserving (CPTP) map for each fixed t.

Standard definitions of one-time Markovianity are related to the notion of *divisibility*. Assume that, for all $t_2 \geq t_1 \geq 0$, there exists an intermediate map $\mathcal{V}(t_2, t_1)$, i.e., a propagator, such that

$$\Lambda_S(t_2) = \mathcal{V}(t_2, t_1) \circ \Lambda_S(t_1); \tag{9}$$

indeed, if $\Lambda_S(t_1)$ is invertible, the propagator can be identified with

$$\mathcal{V}(t_2, t_1) = \Lambda_S(t_2) \circ \Lambda_S(t_1)^{-1}. \tag{10}$$

If each $\mathcal{V}(t_2, t_1)$ can be chosen positive and trace-preserving, the dynamics is *P-divisible* [25, 31, 32]; if, in addition, each $\mathcal{V}(t_2, t_1)$ is completely positive, the dynamics is *CP-divisible* [30]. Both notions have been used to define one-time quantum (non)-Markovianity; in the following, we will focus in particular on the investigation of P-divisibility. This choice is motivated by the comparison with multitime Markovianity: throughout the paper, the latter is assessed via sequential measurement statistics generated by instruments acting on the system alone, and P-divisibility provides a weaker requirement directly testable at the level of system states, whereas CP-divisibility is a stronger condition whose operational meaning involves the presence of an ancilla [38].

B. Two-time correlators and sequential probabilities

We now move to the multitime scenario and introduce the notion of Markovianity we will use throughout our analysis, focusing on sequential protocols where an intervention at t_1 is followed by a readout at t_2, for $0 \le t_1 \le t_2$.

Given an intervention $\mathcal{A}_1$ at time t_1 on the open system, we define the *conditioned two-time propagator*, for $0 \le t_1 \le t_2$,

$$\Phi_{\mathcal{A}_1}(t_2,t_1)[\rho] := \mathrm{Tr}_B\big[e^{(t_2-t_1)\mathcal{L}} \circ (\mathcal{A}_1 \otimes \mathcal{I}_B) \circ e^{t_1\mathcal{L}}[\rho \otimes R]\big], \tag{11}$$

where here and in the following we denote as $\mathcal{I}$ the identity map, and we used the total propagator $e^{t\mathcal{L}}$ defined by Eq.(1). Taking into account also the readout at time t_2, we can further define the two-time quantity

$$F_{\mathcal{A}_2,\mathcal{A}_1}(t_2,t_1) := \mathrm{Tr}_S\left[\mathcal{A}_2 \circ \Phi_{\mathcal{A}_1}(t_2,t_1)[\rho_S(0)]\right], \tag{12}$$

which covers the two main cases of interest for our investigation. Choosing $\mathcal{A}_i = L_{A_i}$ the left multiplication by A_i, i.e., $L_{A_i}[\rho] = A_i\rho$, Eq. (12) yields the time-ordered correlators

$$\langle A_2(t_2)A_1(t_1)\rangle = F_{A_2,A_1}(t_2,t_1) = \mathrm{Tr}_S\left[A_2\Phi_{A_1}(t_2,t_1)[\rho_S(0)]\right], \tag{13}$$

i.e., the two-time correlators associated with the operators A_1 and A_2, which define, for example, linear-response functions and emission spectra [2, 39, 40].

Instead, let $\{\mathcal{M}^{(1)}_{a_1}\}_{a_1\in O_1}$ and $\{\mathcal{M}^{(2)}_{a_2}\}_{a_2\in O_2}$ be instruments at t_1 and t_2, i.e. collections of CP trace-nonincreasing maps such that $\sum_{a_1}\mathcal{M}^{(1)}_{a_1}$ and $\sum_{a_2}\mathcal{M}^{(2)}_{a_2}$ are CPTP, and that describe the statistics of a two-time sequential measurement with outcomes $(a_1,a_2) \in O_1 \times O_2$. The joint probability for outcomes (a_1,a_2) at (t_1,t_2) is obtained by choosing $\mathcal{A}_i = \mathcal{M}^{(i)}_{a_i}$ and applying the trace after the second CP map,

$$\begin{aligned} p(a_2,t_2;\, a_1,t_1) &= F_{\mathcal{M}^{(2)}_{a_2},\mathcal{M}^{(1)}_{a_1}}(t_2,t_1) \\ &= \mathrm{Tr}_S\left[\mathcal{M}^{(2)}_{a_2} \circ \Phi_{\mathcal{M}^{(1)}_{a_1}}(t_2,t_1)[\rho_S(0)]\right]. \end{aligned} \tag{14}$$

From this probability, we can obtain all the information about the statistics of the sequential measurement; for example, the two-time expectation value is given by

$$\overline{A_2(t_2)A_1(t_1)} := \sum_{a_1,a_2} a_1\, a_2\, p(a_2,t_2;\, a_1,t_1). \tag{15}$$

We stress that even when the same quantities A_1 and A_2 are considered, correlators and sequential measurement statistics generally differ because the latter include an explicit state transformation at the intermediate time, whereas the correlator does not. Explicitly, let us consider the two self-adjoint operators $A_1 = A_1^\dagger$ and $A_2 = A_2^\dagger$, with spectral decompositions

$$A_i = \sum_{a_i} a_i \Pi^{(i)}_{a_i}, \quad i = 1,2, \tag{16}$$

with $\Pi^{(i)}_{a_i}$ the projector into the eigenspace of the eigenvalue a_i of the operator A_i, along with the projective measurement associated with it via the Lüders instrument [41],

$$\mathcal{M}^{(i)}_{a_i}[\rho] := \Pi^{(i)}_{a_i}\rho\Pi^{(i)}_{a_i}. \tag{17}$$

Introducing the full-dephasing map associated with the first intermediate measurement,

$$\mathcal{D}_{A_1}[\rho] := \sum_{a_1}\mathcal{M}^{(1)}_{a_1}[\rho] = \sum_{a_1}\Pi^{(1)}_{a_1}\rho\Pi^{(1)}_{a_1}, \tag{18}$$

we can express the discrepancy between the correlator and the sequential-measurement average as

$$\begin{aligned} &\langle A_2(t_2)A_1(t_1)\rangle - \overline{A_2(t_2)A_1(t_1)} \\ &= \mathrm{Tr}_S\left[A_2\, \Phi_{(\mathcal{I}-\mathcal{D}_{A_1})\circ L_{A_1}}(t_2,t_1)[\rho_S(0)]\right]. \end{aligned} \tag{19}$$

This relation shows that the mismatch is entirely due to the off-diagonal (coherence) component $(\mathcal{I} - \mathcal{D}_{A_1})[A_1\rho_S(t_1)]$ removed by the Lüders update, but potentially converted into populations relevant to A_2 by the subsequent dynamics. The sequential statistics depends on the choice of the instrument, and a full agreement with the correlators can occur only for non-invasive interventions or when the coherences in the measurement basis at t_1 vanish or are dynamically irrelevant for the readout measurement at time t_2 [42].

The Markovianity criterion associated with multitime objects we consider consists in the validity of the quantum regression theorem (QRT) [3, 8]. In our notation and restricting to two-time quantities, the latter means that one can perform the replacement

$$\Phi_{\mathcal{A}_1}(t_2,t_1) \mapsto \Lambda_S(t_2-t_1) \circ \mathcal{A}_1 \circ \Lambda_S(t_1), \tag{20}$$

into $F_{\mathcal{A}_2,\mathcal{A}_1}(t_2,t_1)$ defined in Eq.(12), for any initial state $\rho_S(0)$ and readout map $\mathcal{A}_2$. Microscopically, such a replacement is justified when the system-environment correlations built up by the interaction up to time t_1 are irrelevant for the quantities at time t_2 [10]. Indeed, the difference between averages of sequential measurements and two-time correlators manifests itself in the applicability of the QRT, as there are situations where it can be applied to the latter but not to the former, even when the same quantities are involved. In particular, for a two-level system undergoing pure dephasing due to the interaction with a single continuous degree of freedom initially distributed according to a Lorentzian profile, the QRT holds for any two-time correlator [26], while it fails for the average of sequential projective measurements performed in a basis different from the dephasing one [42, 43].

III. BEYOND-QRT TWO-TIME PROPAGATORS

A. Exact expression via projection superoperators

The expression of the two-time expectation values can be rewritten by means of the projection superoperators, to single out in full generality the term leading to a violation of the

QRT.

In particular, introducing the standard projection superoperators [3]

$$\mathcal{P}[\rho] := \mathrm{Tr}_B[\rho] \otimes R, \qquad \mathcal{Q} := \mathcal{I} - \mathcal{P} \tag{21}$$

that act on operators on $S+B$, and replacing the identity $\mathcal{I} = \mathcal{P}+\mathcal{Q}$ in the definition of the conditioned two-time propagator Eq.(11), we can rewrite it as:

$$\begin{aligned}\Phi_{\mathcal{A}_1}(t_2,t_1)[\rho] &= \mathrm{Tr}_B[e^{(t_2-t_1)\mathcal{L}} \circ (\mathcal{A}_1 \otimes \mathcal{I}_B) \\ &\quad \circ (\mathcal{P}+\mathcal{Q}) \circ e^{t_1\mathcal{L}}[\rho \otimes R]\,] \\ &=: \Phi^{(P)}_{\mathcal{A}_1}(t_2,t_1)[\rho] \;+\; \Phi^{(Q)}_{\mathcal{A}_1}(t_2,t_1)[\rho].\end{aligned} \tag{22}$$

Since

$$\mathcal{P}e^{t_1\mathcal{L}}[\rho \otimes R] = \Lambda_S(t_1)[\rho] \otimes R \tag{23}$$

by definition of $\Lambda_S(t)$, we get

$$\begin{aligned}\Phi^{(P)}_{\mathcal{A}_1}(t_2,t_1)[\rho] &= \mathrm{Tr}_B[e^{(t_2-t_1)\mathcal{L}} \circ (\mathcal{A}_1 \otimes \mathcal{I}_B)\; [\Lambda_S(t_1)[\rho] \otimes R]\,] \\ &= \Lambda_S(t_2-t_1) \circ \mathcal{A}_1 \circ \Lambda_S(t_1)[\rho],\end{aligned} \tag{24}$$

i.e., $\Phi^{(P)}_{\mathcal{A}_1}(t_2,t_1)$ exactly corresponds to a term in the form given by the QRT – see Eq.(20). Therefore,

$$\Phi^{(Q)}_{\mathcal{A}_1}(t_2,t_1)[\rho] := \mathrm{Tr}_B[e^{(t_2-t_1)\mathcal{L}} \circ (\mathcal{A}_1 \otimes \mathcal{I}_B) \circ \mathcal{Q} \circ e^{t_1\mathcal{L}}[\rho \otimes R]] \tag{25}$$

is the QRT-violating contribution: it isolates the part of the two-time propagator that is not fixed by the reduced state and the dynamical map only. The size of the possible QRT violation is therefore controlled jointly by (i) how many system-environment correlations and changes in the environmental state are present at time t_1 – i.e., the magnitude of $\mathcal{Q}\rho_{SB}(t_1)$ – and (ii) how strongly the chosen intervention map can convert them into an observable effect for the chosen readout at time t_2. In particular, the back-action of the first measurement and the readout share a nontrivial interplay with the correlation built up by the dynamics: an intervention may amplify QRT deviations by conditioning on bath-correlated degrees of freedom, or suppress them when it reduces precisely those coherences that would feed into the later readout.

The term $\Phi^{(Q)}_{\mathcal{A}_1}$ admits a double-convolution representation

$$\begin{aligned}\Phi^{(Q)}_{\mathcal{A}_1}(t_2,t_1) &= \int_0^{t_2-t_1} d\tau_2 \int_0^{t_1} d\tau_1\; \Lambda_S(t_2-t_1-\tau_2) \\ &\quad \circ \mathcal{K}^{\mathcal{A}_1}(\tau_2,\tau_1) \circ \Lambda_S(t_1-\tau_1),\end{aligned} \tag{26}$$

for the multitime kernel (here and in the following we imply the composition $\circ$ within the kernel)

$$\begin{aligned}\mathcal{K}^{\mathcal{A}_1}(\tau_2,\tau_1)[\rho] &:= \mathrm{Tr}_B[\mathcal{L}_{SB}\, e^{\tau_2\mathcal{Q}\mathcal{L}}\, \mathcal{Q}(\mathcal{A}_1 \otimes \mathcal{I}_B) \\ &\qquad e^{\tau_1\mathcal{Q}\mathcal{L}}\, \mathcal{Q}\mathcal{L}_{SB}\,[\rho \otimes R]],\end{aligned} \tag{27}$$

which can be derived by applying the projection-operator multitime-kernel framework of Ref. [12], developed there for multitime correlation functions. The same technique can be employed identically for a two-time propagator with a generic intermediate intervention $\mathcal{A}_1$, which makes the connection to QRT violations of multitime probabilities explicit.

In fact, focusing on the case where the interventions consist of sequential measurements, we quantify the difference between the exact joint probability distribution p given by Eq.(14) and the joint distribution p_{QRT} obtained by applying the QRT – i.e., replacing Eq.(20) into Eq.(14) – by means of the Kolmogorov distance,

$$\varepsilon_{\mathrm{QRT}}(t_2,t_1) := \frac{1}{2}\sum_{a_1,a_2} \big|\, p(a_2,t_2;\, a_1,t_1) - p_{\mathrm{QRT}}(a_2,t_2;\, a_1,t_1)\big|. \tag{28}$$

Note that since p and p_{QRT} are normalized joint distributions, $\varepsilon_{\mathrm{QRT}}(t_2,t_1) \in [0,1]$, with $\varepsilon_{\mathrm{QRT}} = 0$ if and only if they coincide. Using Eqs.(22) and (24), such a difference can be compactly expressed as

$$\varepsilon_{\mathrm{QRT}}(t_2,t_1) = \frac{1}{2}\sum_{a_1,a_2} \left|\mathrm{Tr}_S\left[\mathcal{M}^{(2)}_{a_2} \circ \Phi^{(Q)}_{\mathcal{M}^{(1)}_{a_1}}(t_2,t_1)[\rho_S(0)]\right]\right|, \tag{29}$$

which directly relates the deviations from the QRT expression in the joint probability distribution to microscopic details of the system-environment interaction.

In the following analysis, we will also consider the t_1-average QRT violation, for a fixed final time $t_2 = t_f$, namely,

$$\bar{\varepsilon}_{\mathrm{QRT}}(t_f) := \frac{1}{t_f}\int_0^{t_f} \varepsilon_{\mathrm{QRT}}(t_f,t_1)\, dt_1. \tag{30}$$

Operationally, $\bar{\varepsilon}_{\mathrm{QRT}}(t_f)$ is the expected QRT error when the first intervention time t_1 is drawn uniformly at random in $[0,t_f]$ and it is indeed more suitable for a comparison with the single-time non-Markovianity.

B. Second-order expression

The explicit evaluation of the two-time quantity in Eq.(12) is in general as complicated as the solution of the global dynamics. In the next Section, we will perform the numerical simulation, in generic parameter regimes, for the spin-boson model we are interested in. Here, instead, we report the lowest order of a perturbative expansion that can be applied to any model in the weak coupling regime.

Since $\mathcal{L}_{SB} = O(\lambda)$, the leading order of the kernel is $O(\lambda^2)$, which is obtained by replacing the $\mathcal{Q}$-projected propagators by their zeroth-order approximation in λ,

$$e^{\mathcal{Q}(\mathcal{L}_0+\mathcal{L}_{SB})\tau} = e^{\mathcal{Q}\mathcal{L}_0\tau} + O(\lambda), \tag{31}$$

so that

$$\begin{aligned}&\mathcal{K}^{\mathcal{A}_1}(\tau_2,\tau_1)[\rho] := \mathrm{Tr}_B[\mathcal{L}_{SB}\, e^{\mathcal{Q}\mathcal{L}_0\tau_2}\, \mathcal{Q}(\mathcal{A}_1 \otimes \mathcal{I}_B) \qquad (32)\\ &e^{\mathcal{Q}\mathcal{L}_0\tau_1}\, \mathcal{Q}\mathcal{L}_{SB}\,[\rho \otimes R]] + O(\lambda^3) =: \tilde{\mathcal{K}}^{\mathcal{A}_1}(\tau_2,\tau_1)[\rho] + O(\lambda^3).\end{aligned}$$

To proceed further, we note that

$$[\mathcal{L}_0,\, \mathcal{Q}] = 0, \tag{33}$$

since, for any $S-B$ operator X,

$$\mathcal{P}\mathcal{L}_0[X] = \mathrm{Tr}_B[\mathcal{L}_0 X]\otimes R = (\mathcal{L}_S[\mathrm{Tr}_B X] + \mathrm{Tr}_B[\mathcal{L}_B X])\otimes R$$
$$= \mathcal{L}_S[\mathrm{Tr}_B X]\otimes R = \mathcal{L}_0[\mathrm{Tr}_B[X]\otimes R] = \mathcal{L}_0\mathcal{P}[X],$$

where we used that $\mathrm{Tr}_B[\mathcal{L}_B X] = 0$ (cyclicity of Tr_B) and $\mathcal{L}_B R = 0$. Exploiting the idempotency relation $Q^2 = Q$ and the power series of the exponential, we then have

$$e^{Q\mathcal{L}_0\tau}QX = e^{\mathcal{L}_0\tau}QX. \tag{34}$$

Moreover, Eq.(5) implies

$$\mathcal{P}\mathcal{L}_{SB}[\rho\otimes R] = -i\lambda\sum_a [S_a,\rho]\ \mathrm{Tr}_B[B_a R]\ \otimes\ R = 0, \tag{35}$$

while, using once more Eq.(33) and $\mathcal{P}(\mathcal{A}_1\otimes\mathcal{I}_B) = (\mathcal{A}_1\otimes\mathcal{I}_B)\mathcal{P}$, one has

$$\mathcal{P}(\mathcal{A}_1\otimes\mathcal{I}_B)e^{\mathcal{L}_0\tau_1}Q[X] = 0. \tag{36}$$

All in all, replacing Eqs.(34)-(36) (along with $Q = \mathcal{I} - \mathcal{P}$) into Eq.(32), we arrive at the following expression of the kernel's leading order:

$$\tilde{\mathcal{K}}^{\mathcal{A}_1}(\tau_2,\tau_1)[\rho] = \mathrm{Tr}_B[\mathcal{L}_{SB}\, e^{\mathcal{L}_0\tau_2}\,(\mathcal{A}_1\otimes\mathcal{I}_B)e^{\mathcal{L}_0\tau_1}\,\mathcal{L}_{SB}\,(\rho\otimes R)], \tag{37}$$

and therefore at the following leading expression of the QRT-violating term:

$$\tilde{\Phi}^{(Q)}_{\mathcal{A}_1}(t_2,t_1) = \int_0^{t_2-t_1} d\tau_2\int_0^{t_1} d\tau_1\ \Lambda_S(t_2-t_1-\tau_2)$$
$$\circ\,\tilde{\mathcal{K}}^{\mathcal{A}_1}(\tau_2,\tau_1)\circ\Lambda_S(t_1-\tau_1). \tag{38}$$

IV. MULTITIME NON-MARKOVIANITY IN THE SPIN-BOSON MODEL

We are now ready to apply the general expressions derived in Sec. III to investigate deviations from the QRT in multi-time probabilities within a paradigmatic open quantum system, namely the spin-boson model [44]. This model provides a simple yet powerful framework to describe the interaction between a two-level system and a dissipative environment, capturing essential features of the evolution in diverse dynamical regimes.

A. The model

We consider the spin–boson Hamiltonian

$$H = H_S + H_B + H_{SB}$$
$$= \frac{\omega_0}{2}\sigma_z + \sum_n \omega_n a_n^\dagger a_n + \sigma_x\otimes\sum_n (g_n a_n + g_n^* a_n^\dagger), \tag{39}$$

i.e. a two-level system linearly coupled to a bosonic environment where σ_i, $i = x,y,z$, are the Pauli matrices of the two-level system, $a_n, a_n^\dagger$ are the annihilation and creation operators of the n-th bosonic mode, ω_0 (ω_n) is the free frequency of the system (n-th mode) and g_n is the coupling between the two-level system and the n-th mode. Assuming an initial global product state, see Eq.(7), and an initial thermal state of the environment at inverse temperature β

$$R = \bigotimes_n \frac{\exp\left[-\beta\,\omega_n a_n^\dagger a_n\right]}{Z_n}, \tag{40}$$

with

$$Z_n = \mathrm{Tr}\left[\exp\left[-\beta\omega_n a_n^\dagger a_n\right]\right], \tag{41}$$

the reduced two-level system dynamics is fixed by the correlation function, see Eq.(6),

$$C(t) = \mathrm{Tr}_B\Big[\sum_{n,m}(g_n e^{-i\omega_n t}a_n + g_n^* e^{i\omega_n t}a_n^\dagger)(g_m a_m + g_m^* a_m^\dagger)\,R\Big]$$
$$= \int_0^{+\infty} d\omega\ J(\omega)\left(e^{-i\omega t}(1+n_\beta(\omega)) + e^{i\omega t}n_\beta(\omega)\right), \tag{42}$$

where we introduced the spectral density [3]

$$J(\omega) = \sum_n |g_n|^2\delta(\omega-\omega_n), \tag{43}$$

and the mean occupation number at inverse temperature β of the n-th mode,

$$n_\beta(\omega_n) := \frac{1}{e^{\beta\omega_n}-1} = \mathrm{Tr}_B\left[a_n^\dagger a_n R\right]. \tag{44}$$

The expression in Eq.(42) allows one to readily take the continuum limit of the bosonic modes, by simply replacing the spectral density defined in (43) with a positive continuous function of ω, for $\omega \geq 0$ (and extending the definition in Eq.(44) to a generic value of $\omega \geq 0$).

In particular, we take into account a Lorentzian spectral density centered at η with width γ [45, 46]:

$$J(\omega) = \frac{2\lambda^2\gamma}{(\omega-\eta)^2+\gamma^2}, \tag{45}$$

which describes, for example, the coupling with a damped cavity mode. In fact, for such a choice of the spectral density, the open-system dynamics can be equivalently obtained using an exact Lindbladian embedding in an enlarged setting $S+M$ with a single damped pseudomode M [36, 37, 47]. Explicitly, the reduced state $\rho_S(t)$ at time t – defined by Eq.(8) – is equal to the reduced state

$$\rho_S(t) = \mathrm{Tr}_M[\rho_{SM}(t)], \tag{46}$$

obtained from the $S+M$ state $\rho_{SM}(t)$ that is the solution of the master equation

$$\frac{d}{dt}\rho_{SM}(t) = \mathcal{L}_{SM}[\rho_{SM}(t)] := -i[H_{\mathrm{SM}},\rho_{SM}(t)]$$
$$+\,2\gamma\,(n_\beta(\eta)+1)\,\mathcal{D}_b[\rho_{SM}(t)] + 2\gamma\,n_\beta(\eta)\,\mathcal{D}_{b^\dagger}[\rho_{SM}(t)]. \tag{47}$$

Here, the coherent system-pseudomode coupling is given by the Hamiltonian

$$H_{\mathrm{SM}} = \frac{\omega_0}{2}\sigma_z + \eta\, b^\dagger b + \lambda\, \sigma_x (b + b^\dagger), \tag{48}$$

where b and $b^\dagger$ are the annihilation and creation operators of the pseudomode, while the dissipator is fixed by the operatorial structure

$$\mathcal{D}_L[\rho] := L\rho L^\dagger - \frac{1}{2}\{L^\dagger L, \rho\}. \tag{49}$$

Moreover, the initial condition is a product state, with a thermal state of the pseudomode, at inverse temperature β:

$$\rho_{SM}(0) = \rho_S(0) \otimes \frac{\exp\left[-\beta\eta b^\dagger b\right]}{\mathrm{Tr}\left[\exp\left[-\beta\eta b^\dagger b\right]\right]}; \tag{50}$$

the case of a zero-temperature environment is indeed recovered for $\rho_M(0) = |0\rangle\langle 0|$ and $n_\beta = 0$. The system-pseudomode evolution is thus fixed by a time-independent master equation, a Gorini-Kossakowski-Lindblad-Sudarshan master equation [48, 49], which means that the $S+M$ compound evolves in a Markovian (time-homogeneous) way; crucially, the reduced two-level system evolution will generally be non-Markovian: all the memory effects are encompassed in the interaction with the pseudomode. Even more, such a picture concerns not only the dynamics, but also the multitime expectation values – being them correlators or sequential probabilities: Eq.(12) can be evaluated exactly by means of the QRT-expression on the enlarged $S+M$ system [50, 51], according to

$$F_{\mathcal{A}_2,\mathcal{A}_1}(t_2,t_1) = \mathrm{Tr}_{SM}\Big[(\mathcal{A}_2 \otimes \mathcal{I}_M) \circ e^{(t_2-t_1)\mathcal{L}_{SM}} \circ (\mathcal{A}_1 \otimes \mathcal{I}_M) \circ e^{t_1\mathcal{L}_{SM}}[\rho_{SM}(0)]\Big]; \tag{51}$$

indeed, this expression includes all the violations of the QRT with respect to the reduced S dynamical map.

Numerically, to solve the master equation (47) and the two-time expression (51), we truncate the pseudomode Hilbert space to a finite Fock cutoff $n_{\max}$ and verify convergence of all reported quantities upon increasing $n_{\max}$. In particular, we work in the column-stacking (Liouville) representation and exploit QuTiP libraries [52, 53]. The dynamics of the enlarged $S+M$ system is propagated through the exponential of the corresponding Lindblad generator, while multitime probabilities are computed by applying the relevant intervention maps on the system at the prescribed intermediate times, retaining the full $S+M$ state across the intervention, and propagating to the subsequent readout time under the same generator.

B. Two-time non-Markovianity

The focus of our investigation is the characterization of the multitime probabilities emerging from sequential measurements in general, non-Markovian, regimes where the QRT cannot be applied. In particular, as explained in Sec.III A, we use $\varepsilon_{\mathrm{QRT}}(t_2,t_1)$ defined in Eq.(28) to quantify the deviations from the QRT and then the multitime non-Markovianity.

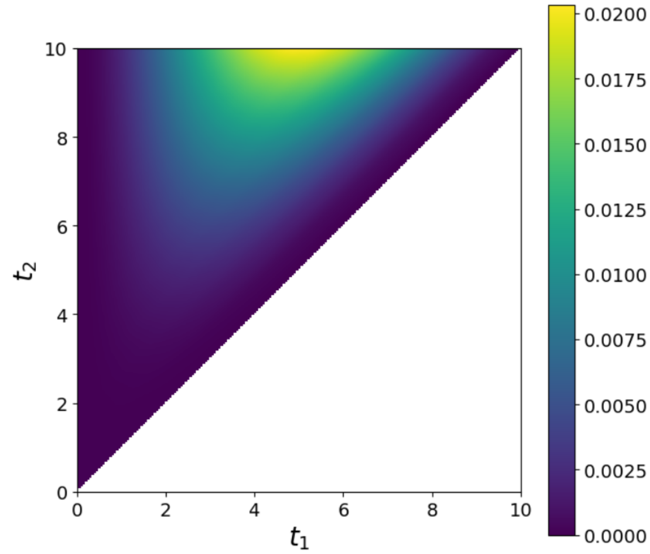

Figure 1. Two-time landscape of QRT violation. Heatmap of the QRT-violation quantifier $\varepsilon_{\mathrm{QRT}}(t_2,t_1)$ defined in Eq.(28 over the (t_1,t_2) plane, for $\gamma = 0.1$, $\omega_0 = \eta = 4.5$, $\lambda = 0.1$ [here and in the following all parameters are in arbitrary units], $\rho_S(0) = |0\rangle\langle 0|$ and a zero-temperature environment; the pseudomode levels are cut at $n_{max} = 8$. The white triangular region corresponds to $t_2 < t_1$.

In Fig. 1 we report $\varepsilon_{\mathrm{QRT}}(t_2,t_1)$ over the (t_1,t_2) plane for a sequential projective measurement protocol in the σ_z basis at both times. We observe that the QRT error is smallest close to the diagonal $t_2 \simeq t_1$ and increases for larger temporal separations, indicating that the impact of system–bath correlations across the intervention at t_1 becomes more pronounced as the second readout is delayed. Moreover, the dependence on t_1 is nontrivial: for fixed t_2 there is typically a window of intermediate t_1 values where $\varepsilon_{\mathrm{QRT}}$ is maximized, reflecting the fact that the same intervention can be more or less disruptive depending on how much correlation has built up between system and environment prior to t_1.

To better understand the non-Markovianity of the two-time statistics, we compare it with the non-Markovianity of the dynamics. As stated in Sec.II A, we identify the latter with a violation of the P-divisibility of the dynamics, i.e., the non-positivity of the propagator $\mathcal{V}(t_2,t_1)$ in Eq.(9). Since we are dealing with a two-dimensional open quantum system, we can take advantage of a direct characterization of the dynamical map and propagators associated with it. In fact, any two-level quantum system state is univocally associated with a Bloch vector $\mathbf{r} \in \mathbb{R}^3$, $\|\mathbf{r}\| \leq 1$, so that any map $\Lambda : \rho \mapsto \Lambda[\rho]$ is associated with an affine transformation [54],

$$\mathbf{r} \mapsto \mathbf{r}' = M\,\mathbf{r} + \mathbf{v}, \tag{52}$$

where M is a 3×3 real matrix and $\mathbf{v} \in \mathbb{R}^3$. A two-level system map is trace preserving and positive if and only if it maps the Bloch ball into itself, so that positivity of Λ is equivalent to the requirement

$$\max_{\|\mathbf{r}\|\leq 1} \|M\mathbf{r} + \mathbf{v}\| \leq 1. \tag{53}$$

Thus, we quantify the non-positivity of the propagator $\mathcal{V}(t_2, t_1)$ as

$$q(t_2, t_1) := \max\left\{0, \max_{\mathbf{r}\in\mathcal{S}} \|M(t_2, t_1)\mathbf{r} + \mathbf{v}(t_2, t_1)\| - 1\right\}, \quad (54)$$

where $M(t_2, t_1)$ and $\mathbf{v}(t_2, t_1)$ denote the affine representation of $\mathcal{V}(t_2, t_1)$ according to Eq. (52), and we estimate the maximum by sampling a set of directions $\mathcal{S} = \{\mathbf{r}_\ell\}$ on the unit sphere using Fibonacci sampling [55, 56]. In addition, to facilitate the comparison between two-time and one-time non-Markovianity, we consider the time-average quantifier of QRT violation, $\bar{\varepsilon}_{\rm QRT}(t_f)$ [see Eq. (30)], and introduce the corresponding average quantifier for the one-time case,

$$\bar{N}(\Lambda; t_f) := \frac{1}{t_f}\int_0^{t_f} dt_1\, q(t_f, t_1). \quad (55)$$

The left panel of Fig. 2 reports the average two-time non-Markovianity quantifier $\bar{\varepsilon}_{\rm QRT}(t_f)$ over the (ω_0, γ) plane at fixed bath peak η (and fixed t_f, λ, and measurement protocol). The plot highlights a region where QRT yields appreciable errors in sequential statistics. As expected, the largest values occur for long bath memory (small γ), while in the short-memory regime (large γ) the QRT error becomes negligible over the explored range of ω_0. The average one-time non-Markovianity, as quantified by $\bar{N}(\Lambda; t_f)$, is shown in the central and right panels of Fig. 2. The central panel is characterized by the same parameter values as the left panel, and the comparison between the two panels illustrates that one-time and two-time non-Markovianity are related but not equivalent: while both quantifiers tend to increase in long-memory regimes, their detailed dependence on (ω_0, γ) differs. Operationally, $\bar{\varepsilon}_{\rm QRT}(t_f)$ probes multitime, protocol-dependent, memory across the intervention at t_1, whereas $\bar{N}(\Lambda; t_f)$ is constructed solely from properties of the one-time map $\Lambda_S(t)$. The P-divisibility violation $\bar{N}(\Lambda; t_f)$ is confined to a narrow strip at small γ and exhibits two off-resonant maxima, with a local minimum at $\omega_0 = \eta$. By contrast, the QRT violation $\bar{\varepsilon}_{\rm QRT}(t_f)$ displays a single peak centered at resonance and extends to substantially larger values of γ. This shows that the two quantifiers probe different aspects of memory. The quantity $\bar{N}(\Lambda; t_f)$ is determined solely by the behavior of the reduced propagator and, in particular, by the loss of positivity of the intermediate map $\mathcal{V}(t_f, t_1)$, indicating a back flow of information to the open system during the time interval $[t_1, t_f]$. The quantity $\bar{\varepsilon}_{\rm QRT}(t_f)$, instead, is sensitive to the system–environment correlations generated up to time t_1 and that subsequently impact the measured sequential statistics. For the protocol considered here, this effect is strongest near resonance, where the system and pseudomode are most efficiently coupled, and thus stronger correlations are built up. Accordingly, there are parameter regions in which no appreciable deviation from *P*-divisibility is detected within the chosen observation window, while the sequential statistics still display a clear breakdown of the QRT factorization. This provides a direct illustration that operational multitime memory is not determined by one-time divisibility properties of the reduced dynamics alone.

The double-peak structure observed in $\bar{N}(\Lambda; t_f)$ at $t_f = 10$ is consistent with a transient feature within the chosen observation window. Near resonance, where the system and pseudomode are most efficiently coupled, the signature of an information backflow in the reduced propagator appears to build up on a comparatively longer timescale within the chosen observation window. At finite detuning, an earlier visible backflow contribution can instead emerge, giving rise to the two lateral maxima. For larger t_f, these transient differences become substantially less pronounced, indicating that the off-resonant enhancement is largely a feature of the chosen observation window. This trend is consistent with the right panel of Fig. 2, where $\bar{N}(\Lambda; t_f)$ is reported for a larger value of t_f and the double-peak structure is strongly reduced. By contrast, $\bar{\varepsilon}_{\rm QRT}(t_f)$ already exhibits a single resonance-centered peak at shorter t_f, suggesting that the breakdown of the QRT is governed by a shorter timescale, associated more directly with the buildup of system–environment correlations than with the completion of a full oscillation cycle of the information exchange between the system and the environment.

Perturbative analysis: improvement from the second-order correction.

The benchmark with the exact two-time statistics provided by the system plus pseudomode configuration further allows us to check the quality of the second-order correction to the QRT we derived in Sec. III B. In particular, we compare the exact two-time probability $p(a_2, t_2;\, a_1, t_1)$, evaluated via Eq.(51), and the approximated two-time probability $p_{\lambda^2}(a_2, t_2;\, a_1, t_1)$, which includes the second-order correction to the QRT given by Eq.(38), i.e., that is obtained by replacing the latter, along with Eqs.(22), (24) into the definition of the two-time probability, Eq.(14), so that

$$p_{\lambda^2}(a_2, t_2;\, a_1, t_1) = p_{\rm QRT}(a_2, t_2;\, a_1, t_1) + \mathrm{Tr}_S\left[\mathcal{M}^{(2)}_{a_2} \circ \tilde{\Phi}^{(Q)}_{\mathcal{M}^{(1)}_{a_1}}(t_2, t_1)[\rho_S(0)]\right]. \quad (56)$$

To quantify the difference between $p(a_2, t_2;\, a_1, t_1)$ and $p_{\lambda^2}(a_2, t_2;\, a_1, t_1)$, we use once again the Kolmogorov distance [compare with Eq.(28)], namely,

$$\varepsilon_{\lambda^2}(t_2, t_1) := \frac{1}{2}\sum_{a_1, a_2}\left| p(a_2, t_2;\, a_1, t_1) - p_{\lambda^2}(a_2, t_2;\, a_1, t_1)\right|. \quad (57)$$

Figure 3 shows that the second-order correction yields a substantially smaller residual discrepancy, compared to the whole QRT violation $\varepsilon_{\rm QRT}(t_2, t_1)$, confirming that the leading memory term captured by Eq. (37) significantly moves the sequential statistics in the correct direction; indeed, the residual error grows as the coupling strength increases, remaining anyway below 10^{-3} in the considered parameter regime.

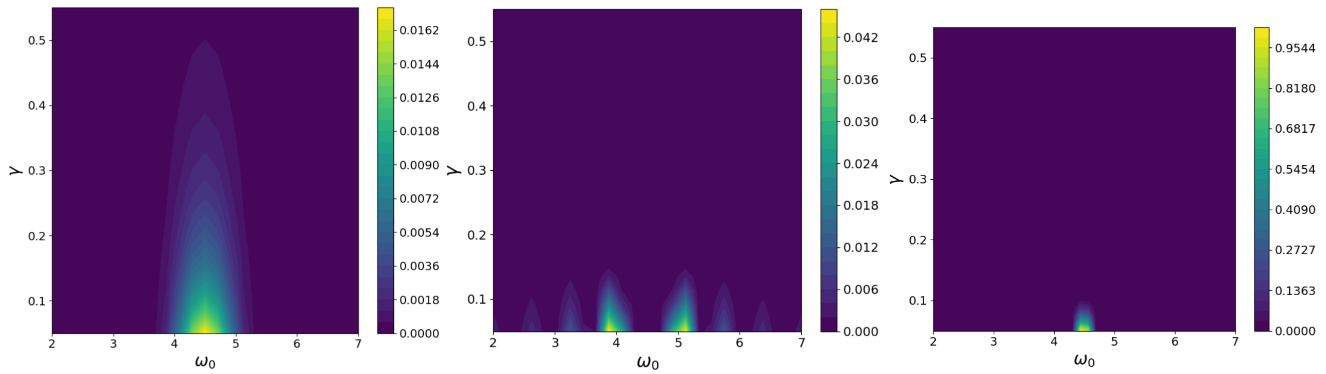

Figure 2. Average two-time non-Markovianity versus average one-time non-Markovianity. (Left) $\bar{\varepsilon}_{\mathrm{QRT}}(t_f)$ defined in Eq.(30) over the (ω_0, γ) plane, for the sequential-measurement protocol with $\rho_S(0) = |0\rangle\langle 0|$ and projective σ_z measurements at t_1 and $t_f = 10$; $\eta = 4.5, \lambda = 0.1, n_{max} = 10$ and zero-temperature environment. (Central) $\bar{N}(\Lambda; t_f)$ defined in Eq.(55) for the corresponding one-time reduced dynamics. (Right) Same quantity and values as the central panel, but for $t_f = 30$.

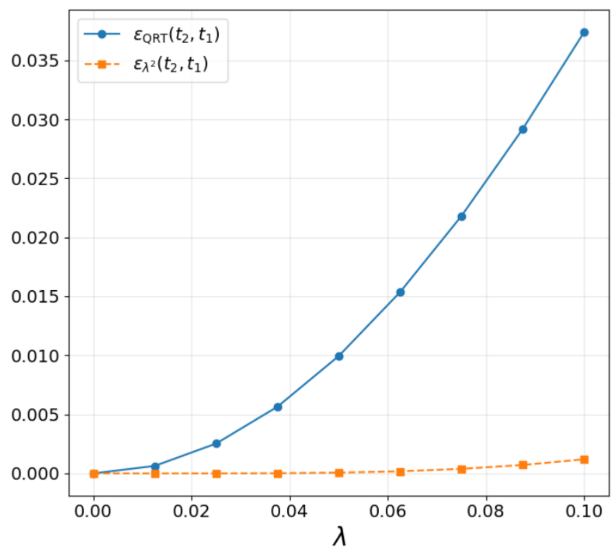

Figure 3. Second-order correction beyond-QRT. QRT violation $\varepsilon_{\mathrm{QRT}}(t_2, t_1)$ as defined in Eq.(28) and residual error after including the second-order correction, $\varepsilon_{\lambda^2}(t_2, t_1)$ [Eq.(57)], as a function of the coupling strength λ. Parameters: $\gamma = 0.1$, $\omega_0 = \eta = 4.5$, zero-temperature environment and $n_{max} = 8$. Protocol: $\rho_S(0) = |+\rangle\langle +|$; Lüders projective measurement in the σ_z basis at $t_1 = 5$ and readout in the σ_x basis at $t_2 = 10$.

C. Protocol dependence: temperature, initial states and measurement bases

We now analyze how the average QRT violation depends on the details of the multitime protocol, namely, the initial state, environmental temperature and the choice of the sequential measurements.

First, we repeat the scan over the parameter space (ω_0, γ) at fixed final time $t_2 = t_f$, for different values of β; moreover, we fix the protocol to sequential projective measurements in the σ_z basis, with a Lüders instrument, and the initial state $\rho_S(0) = |0\rangle\langle 0|$.

Figure 4 shows $\bar{\varepsilon}_{\mathrm{QRT}}$ for representative inverse temperatures β, and hence thermal occupations n_β. Across all cases, the region where QRT deviations are appreciable remains localized in the long-memory regime (small γ) and is typically enhanced near resonance $\omega_0 \simeq \eta$. Increasing the temperature leads to a quantitative enhancement of the average QRT violation in the same long-memory region, indicating that thermal fluctuations amplify the two-time memory for the present spin–boson setting. This behavior can be understood in terms of the $\mathcal{P} - \mathcal{Q}$ decomposition introduced in Sec. III A. At finite temperature the pseudomode is thermally populated, which increases the number of pathways through which system–mode correlations can build up prior to intervention at time t_1. As a consequence, the $\mathcal{Q}$-component of the joint state at t_1—the sole source of QRT violation in Eq. (25)—can have a stronger impact on the observed subsequent statistics.

We next consider the dependence on the specific measurement protocol: The quantity $\bar{\varepsilon}_{\mathrm{QRT}}(t_f)$ is defined in terms of joint outcome probabilities and therefore depends on both the initial state and the measurement scheme. In particular, the initial preparation determines how correlations develop up to time t_1, while the choice of measurement basis affects both the back-action at t_1 and the components of the state probed at the final time. To assess these effects, we repeat the parameter scan for two initial states, $|0\rangle$ and $|+\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$, and for projective measurements in either the σ_z or σ_x basis (using the same basis at both times). The results, shown in Fig. 5, indicate that the magnitude of the average QRT error depends on the protocol: changing either the initial state or the measurement basis can suppress or enhance $\bar{\varepsilon}_{\mathrm{QRT}}(t_f)$, consistently with the fact that different protocols are sensitive to different components of the memory due to the system-environment interaction. At the same time, the location of the region where QRT violations are largest is comparatively robust: across all panels, it remains concentrated in the long-memory regime (small γ) and near resonance $\omega_0 \simeq \eta$.

The behavior in Fig. 5 can be qualitatively understood from the relation between the measurement basis and the system–bath coupling. In the present model, the interaction is

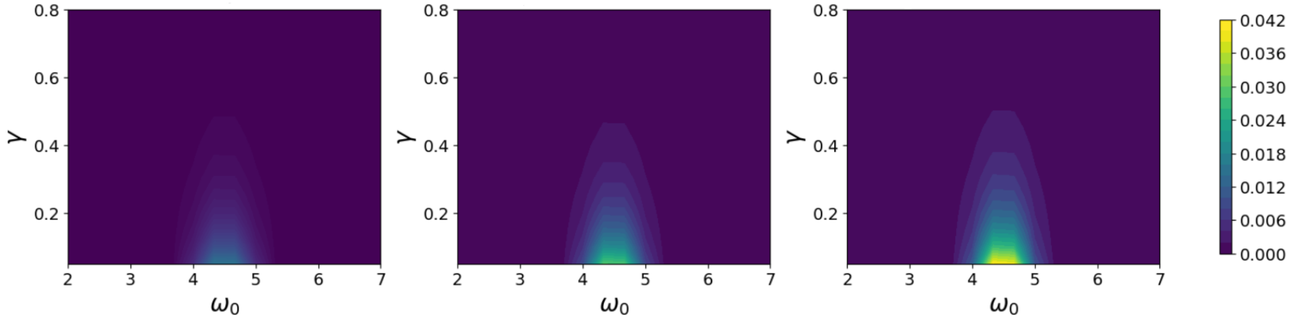

Figure 4. Temperature dependence of the average QRT violation. $\bar{\varepsilon}_{\mathrm{QRT}}(t_f)$ as defined in Eq.(30) over the (ω_0, γ) plane at fixed bath peak $\eta = 4.5$ and coupling $\lambda = 0.1$, using the sequential measurement protocol consisting of $\rho_S(0) = |0\rangle\langle 0|$ and projective σ_z measurements with Lüders update. The three panels refer to different environmental temperatures: zero temperature (left panel), $\beta = 0.25$ (central panel), corresponding to a mean occupation number $n_\beta = 0.48$, and $\beta = 0.154$ (right panel), corresponding to a mean occupation number $n_\beta = 1.002$; the pseudomode has been correspondingly truncated at different values of the maximum level n_{max}: 8 (left), 13 (center) and 20 (right).

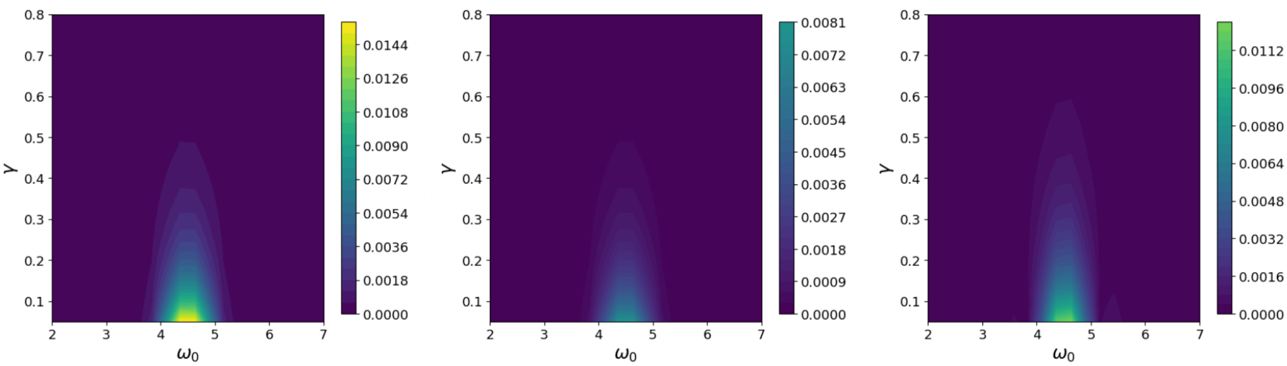

Figure 5. Protocol dependence of the average QRT violation. $\bar{\varepsilon}_{\mathrm{QRT}}(t_f)$ over the (ω_0, γ) plane for three sequential measurement protocols. Left: σ_z basis, $\rho_S(0) = |0\rangle\langle 0|$ (reference protocol, same as Fig. 2 left). Center: σ_z basis, $\rho_S(0) = |+\rangle\langle +|$ (different initial state). Right: σ_x basis, $\rho_S(0) = |0\rangle\langle 0|$ (different measurement basis). All other parameters are the same as in Fig. 2.

proportional to σ_x, so that states aligned with the σ_x basis experience reduced fluctuations of the coupling operator and tend to build weaker correlations with the environment before the intervention, which leads to smaller deviations from the QRT. Conversely, preparations and measurements in a basis that does not commute with the coupling operator, such as σ_z, generally enhances these fluctuations and make the sequential statistics more sensitive to system-environment correlations, resulting in larger QRT violations.

D. Higher-order sequential-measurement memory: three-time QRT violations

The two-time analysis conducted until now probes the QRT across a single intervention in the course of the evolution. It is interesting, however, to look at the impact of a larger number of intermediate-time interventions, which is what we are going to do now by comparing the two-time QRT error with its three-time analogue.

Consider three interventions at $0 \le t_1 \le t_2 \le t_3$ described by the instruments $\{\mathcal{M}^{(j)}_{x_j}\}_{x_j}$ (CP, trace non-increasing maps – see Sec.II B). Generalizing Eq. (14), the exact joint distribution can be computed from the pseudomode embedding by propagating ρ_{SM} and applying the corresponding instruments on S at each intervention time:

$$p(x_3, t_3;\ x_2, t_2;\ x_1, t_1) = \mathrm{Tr}_{SM}\Big[(\mathcal{M}^{(3)}_{x_3} \otimes \mathcal{I}_M) \circ e^{(t_3-t_2)\mathcal{L}_{SM}} \circ (\mathcal{M}^{(2)}_{x_2} \otimes \mathcal{I}_M) \circ e^{(t_2-t_1)\mathcal{L}_{SM}} \circ (\mathcal{M}^{(1)}_{x_1} \otimes \mathcal{I}_M) \circ e^{t_1\mathcal{L}_{SM}}[\rho_{SM}(0)]\Big]. \tag{58}$$

The QRT prediction generalizes indeed the two-time expression – see Eqs.(14) and (20) – thus providing

$$p_{\mathrm{QRT}}(x_3, t_3;\ x_2, t_2;\ x_1, t_1) = \mathrm{Tr}_S\big[\mathcal{M}^{(3)}_{x_3} \circ \Lambda_S(t_3 - t_2) \circ \mathcal{M}^{(2)}_{x_2} \circ \Lambda_S(t_2 - t_1) \circ \mathcal{M}^{(1)}_{x_1} \circ \Lambda_S(t_1)[\rho_S(0)]\big], \tag{59}$$

i.e. it concatenates the same one-time reduced map $\Lambda_S(\cdot)$ between successive interventions. Also here, we quantify the three-time QRT error by the Kolmogorov distance

$$\varepsilon^{(3)}_{\mathrm{QRT}}(t_3, t_2, t_1) = \frac{1}{2} \sum_{x_1, x_2, x_3} \Big| p(x_3, t_3;\ x_2, t_2;\ x_1, t_1) - p_{\mathrm{QRT}}(x_3, t_3;\ x_2, t_2;\ x_1, t_1) \Big|, \tag{60}$$

and define the average version corresponding to uniform sampling [compare with Eq.(30)],

$$\bar{\varepsilon}^{(3)}_{\rm QRT}(t_f) := \frac{1}{t_f}\int_0^{t_f} dt_2\, \frac{1}{t_2}\int_0^{t_2} \varepsilon^{(3)}_{\rm QRT}(t_f, t_2, t_1)\, dt_1; \tag{61}$$

more explicitly, Eq. (61) is the expected three-time QRT error for fixed $t_3 = t_f$, when t_2 is drawn uniformly in $[0, t_f]$ and, conditional on t_2, the earlier time t_1 is drawn uniformly in $[0, t_2]$.

Figure 6 compares the average two-time error $\bar{\varepsilon}_{\rm QRT}(t_f)$ and the average three-time error $\bar{\varepsilon}^{(3)}_{\rm QRT}(t_f)$ for different measurement protocols: commuting (upper row) and non-commuting (lower row) projective measurements at the intermediate times. In both cases, the three-time error peaks in the same parameter region where the two-time error is maximal and is overall larger, indicating that adding an intermediate intervention tends to make the QRT approximation worse, especially where the environment retains memory. Furthermore, by comparing the two different protocols we note that for non-commuting measurement protocols the two-time QRT error can be significantly reduced, while the corresponding three-time deviation is enhanced, with respect to the commuting measurements protocol. This represents a genuinely multitime manifestation of memory, where the number of intermediate interventions does influence how the statistics depart from the Markovian description. More specifically, in the non-commuting sequence the intermediate measurement in the x basis constitutes a non-commuting intervention with respect to the z preparation, the first measurement and the readout measurement. The back-action of the x-basis measurement can modify the joint system–bath state in a way that enhances the contribution of the memory-carrying component (the $\mathcal{Q}$-part in the $\mathcal{P}$–$\mathcal{Q}$ decomposition) to the subsequent evolution. The final measurement in the z basis then acts as a readout that converts these correlations into observable differences in the three-time probabilities. This mechanism is not present when the x-basis measurement coincides with the final readout, since in that case the correlations do not propagate further into a subsequent measurement step. In this sense, the choice of protocol determines which temporal order of the statistics is required to reveal non-Markovian features: a two-time probability may remain close to the QRT prediction, while deviations become visible only at the level of three-time statistics.

V. CONCLUSIONS AND OUTLOOK

We have analyzed multitime memory effects in open quantum systems through deviations from the QRT in sequential measurement statistics. At the structural level, we showed that the two-time propagators and probabilities can be decomposed into a QRT-like contribution, fully determined by the reduced dynamical map, and a complementary term that accounts for system–environment correlations across the intervention and thus the subsequent memory effects. This provides an explicit characterization of the part of the sequential dynamics that is not captured by one-time reduced evolution.

In the weak-coupling regime, we derived a second-order correction to the QRT, expressed in terms of the reduced map and bath correlation functions. In the spin–boson model, this correction significantly improves the prediction of sequential statistics compared to the standard QRT approximation. Framing the analysis in terms of quantum instruments, we also introduced an operational quantifier of QRT breakdown based on the distance between exact and QRT-predicted joint probabilities.

The results highlight that multitime memory, as probed through sequential statistics, depends on the specific measurement protocol and is not fully characterized by one-time properties of the reduced dynamics, such as divisibility. In particular, different choices of initial state and measurement basis can lead to different sensitivities to system–environment correlations. Moreover, higher-order sequential statistics can reveal memory effects that remain small at the level of a given two-time protocol.

Possible extensions of our analysis include the systematic investigation of higher-order temporal protocols, stronger-coupling corrections beyond the leading order term, and the treatment of initially correlated system–environment states. It would also be interesting to connect the present framework to broader notions of temporal nonclassicality formulated directly at the level of multitime probability distributions, for example through violations of Kolmogorov consistency conditions in sequential measurement protocols [23, 57–60].

ACKNOWLEDGMENTS

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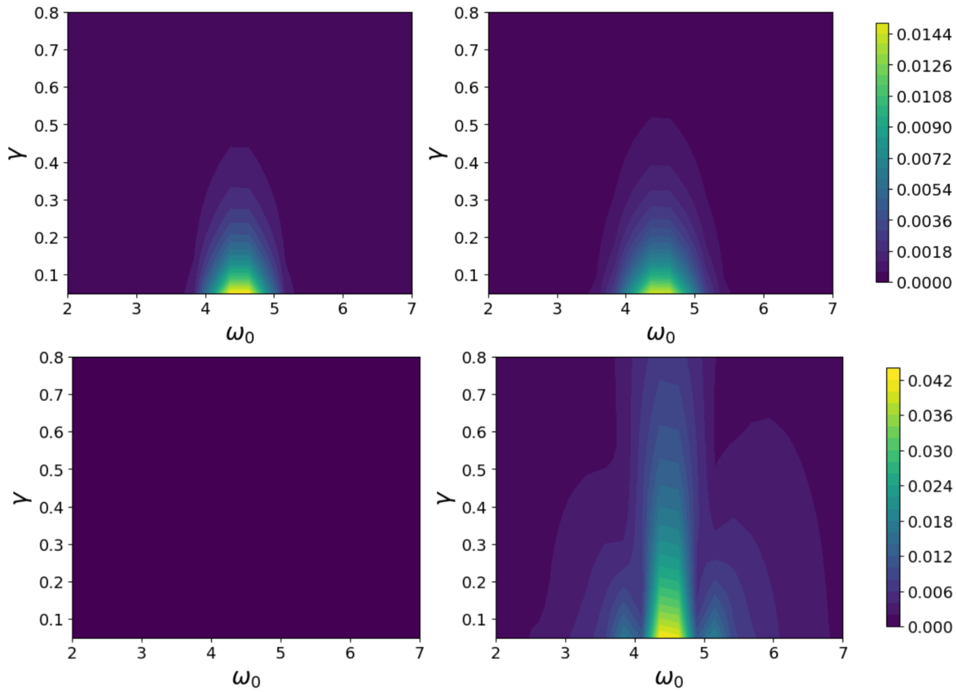

Figure 6. Two- vs three-time QRT error. Left panels: two-time average QRT violation $\bar{\varepsilon}_{\mathrm{QRT}}(t_f)$ [Eq. (30)]. Right panels: three-time average QRT violation $\bar{\varepsilon}^{(3)}_{\mathrm{QRT}}(t_f)$ [Eq. (61)]. Both quantities are shown over the (ω_0, γ) plane. The upper row corresponds to a commuting protocol with projective σ_z measurements at all times. The lower row corresponds to a non-commuting protocol: (σ_z, σ_x) in the two-time case (left) and $(\sigma_z, \sigma_x, \sigma_z)$ in the three-time case (right). Initial state $\rho_S(0) = |0\rangle\langle 0|$, zero-temperature environment, $\eta = 4.5$, $\lambda = 0.1$, $n_{max} = 8$, $t_f = 10.0$.

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